FOURTH ANNUAL ECC
UNDERGRADUATE
MATHEMATICS COMPETITION

March 31, 2001

Time control: Three hours

To the team members: The problems are meant to be fun as well as challenging. You should probably not expect to work all ten of them in the allotted time. Each problem counts 10 points. Partial credit will be given for significant progress or for significant partial solutions, but a thorough job on a few of them will probably be better than some exploratory work on all.

NO BOOKS, NOTES, CALCULATORS, COMPUTERS OR NON-TEAM-MEMBERS may be consulted.

Each team may submit one solution to each problem. Think of your solution as an essay; a logical argument which makes clear why your answer to the question is correct, or why the assertion whose proof is called for in the problem is true.

PLEASE BEGIN EACH PROBLEM ON A NEW SHEET OF PAPER. Team identification and problem number should be clearly given at the top of each sheet of paper submitted.

1. The area is 9.

If $m > 0$ and the area of the region enclosed by the graphs of $y = mx$ and $y = x^2$ is 9, what is the value of m?

2. Limit of a fraction.

Given that the function f is differentiable and that $\lim_{x \to 0} f(x) = 2$, determine

$$\lim_{x \to 0} \frac{e^{3x} - 1}{xf(x)}.$$

3. Divisible by 48 for odd n.

Prove that for all odd integers n,

$$F(n) = n^4 - 4n^2 + 3$$

is divisible by 48.

4. Greater than or equal to 2.

Suppose that a and b are nonzero real numbers. Prove that

$$\left| \frac{a}{b} + \frac{b}{a} \right| \geq 2.$$

5. A probability.

Numbers in the interval $(0, 1)$ are generated by a device according to a uniform distribution. This means that for every subinterval (a, b) of (0,1), the probability that the number generated in any given instance falls in the interval (a, b) is $b - a$. Thus, e.g., whenever a number is generated, the probability that it falls in the interval $(.25, .30)$ is 0.05; the probability that it falls in the interval $(.75, 1)$ is 0.25.

Suppose five numbers are generated independently in succession. What is the probability that the second-largest of them is greater than 0.6? Express your answer as an exact fraction in lowest terms.

6. Sum the series.

Find the sum of the series

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)(n+2)},$$

and justify your answer.

7. What is $f(-1)$?.

The function f is real-valued on a certain open interval containing -1 and 0, and has a continuous derivative satisfying $f'(x) = (x + f(x))^2$ and $f(0) = 1$. Find $f(-1)$ (and justify your answer).

8. Which is larger?.

Determine, with proof, which of the rational numbers

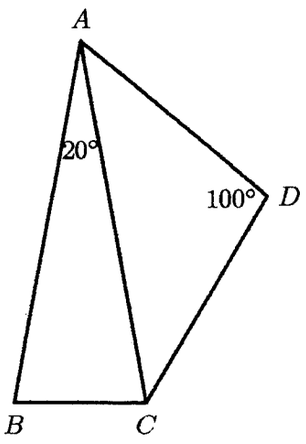
$$a = \frac{2001^{2000} + 1}{2001^{2001} + 1}$$

and

$$b = \frac{2001^{2001} + 1}{2001^{2002} + 1}$$

is larger.

9. An equality among side lengths.



In the figure at the left, sides AB and AC are equal in length, as are sides AD and CD. Angle BAC is 20° and angle ADC is 100°. Prove that $\overline{AB} = \overline{BC} + \overline{CD}$. Here $\overline{AB}$ denotes the length of side AB

10. A sequence of polynomials.

The sequence of polynomials $\{P_n(x)\}$ is defined as follows: $P_0(x) = 1$, $P_1(x) = x^2$, and

$$P_{n+1}(x) = (x^2 - 2)P_n(x) - P_{n-1}(x) + 2. \tag{1}$$

Note that the recursion (1) may be used to extend the sequence in both directions, and $P_n(x)$ is regarded as being defined for all integers n. Prove that for all integers n,

$$(P_n(x) - 1)^2 = P_{n+1}(x)P_{n-1}(x). \tag{2}$$